# LUNG COMPLIANCE AND PULMONARY SURFACTANT

BY

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# LEARNING OBJECTIVES

By the end of the lecture, the students should be able to:

- Define Lung compliance
- Illustrate Compliance diagram of lungs
- Explain how do lungs adapt and why?
- Describe tension on lung surface
- Explain lung and chest compliance
- Define Laplace law
- Applied

# LUNG COMPLIANCE

The extent to which the lungs will expand for each unit increase in transpulmonary pressure (if enough time is allowed to reach equilibrium) is called the lung compliance.

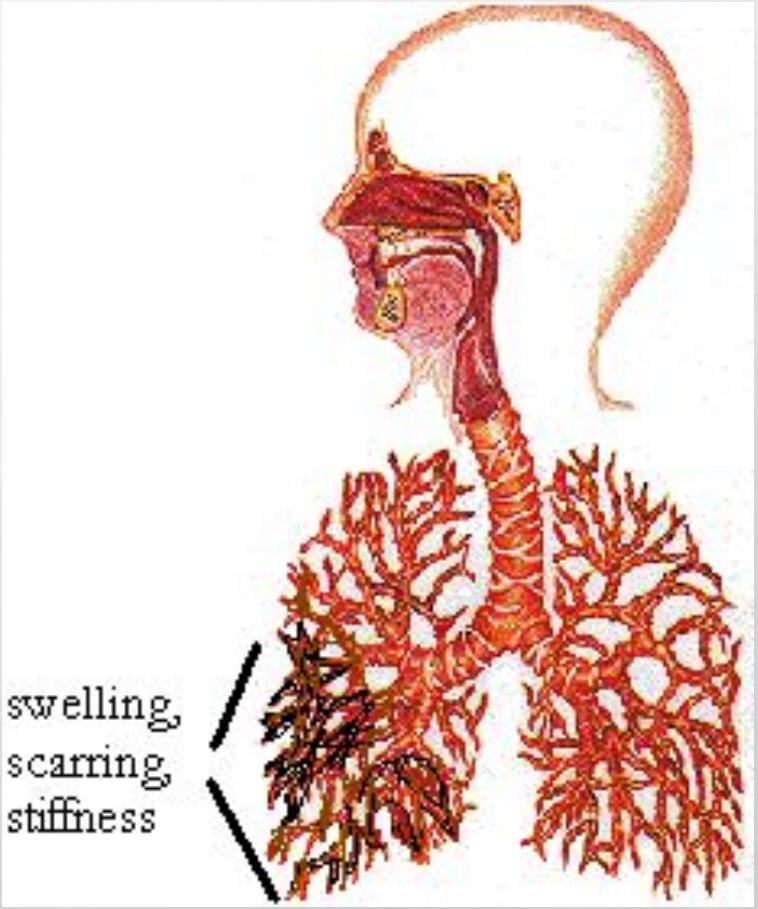
The total compliance of both lungs together in a normal adult human averages about 200 mls of air per cm of water transpulmonary pressure.

# LUNG COMPLIANCE

- Change in lung volume for each unit change in transpulmonary pressure is called stretchiness or expansibility of lungs

**Significance of Determining Compliance:**

Determination of compliance is useful as it is the measure of stiffness of lungs. Stiffer the lungs, less is the compliance.



# COMPLIANCE DIAGRAM OF LUNGS

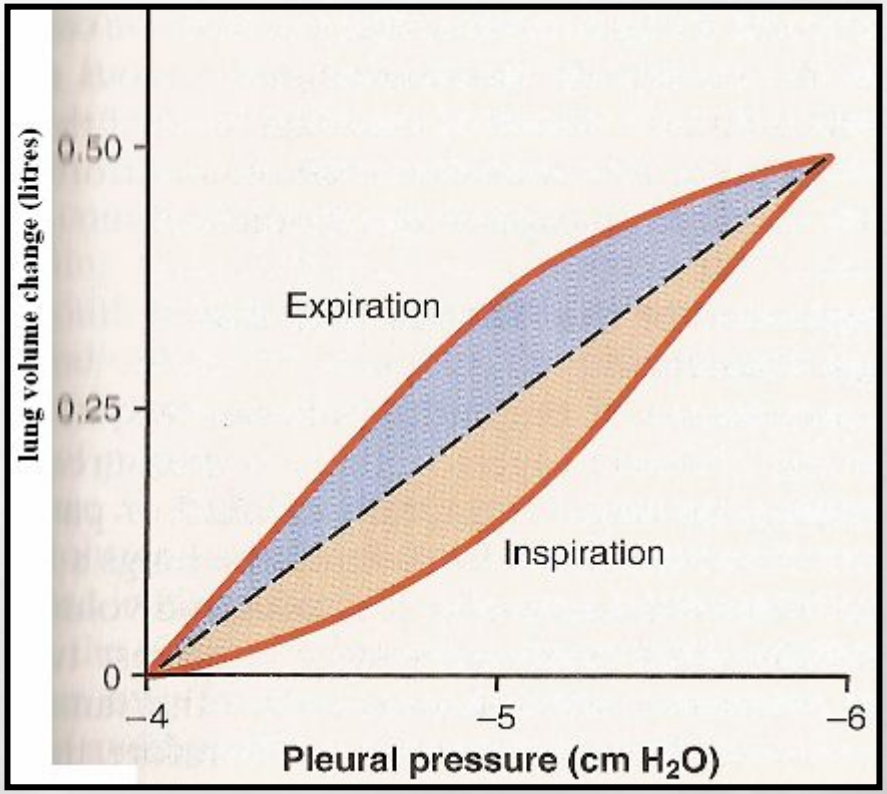
There are 2 different curves according to different phases of respiration.

The curves are called:

- Inspiratory compliance curve
- Expiratory compliance curve

Shows the capacity of lungs to "adapt" to small changes of transpulmonary pressure. Compliance is seen at low volumes (because of difficulty with initial lung inflation) and at high volumes (because of the limit of chest wall expansion).

The total work of breathing of the cycle is the area contained in the loop.



# LUNG COMPLIANCE IN RELATION TO ELASTIC FORCES

COMPLIANCE OF LUNGS OCCUR DUE TO ELASTIC FORCES:

ELASTIC FORCES OF THE LUNG PARENCHYMA

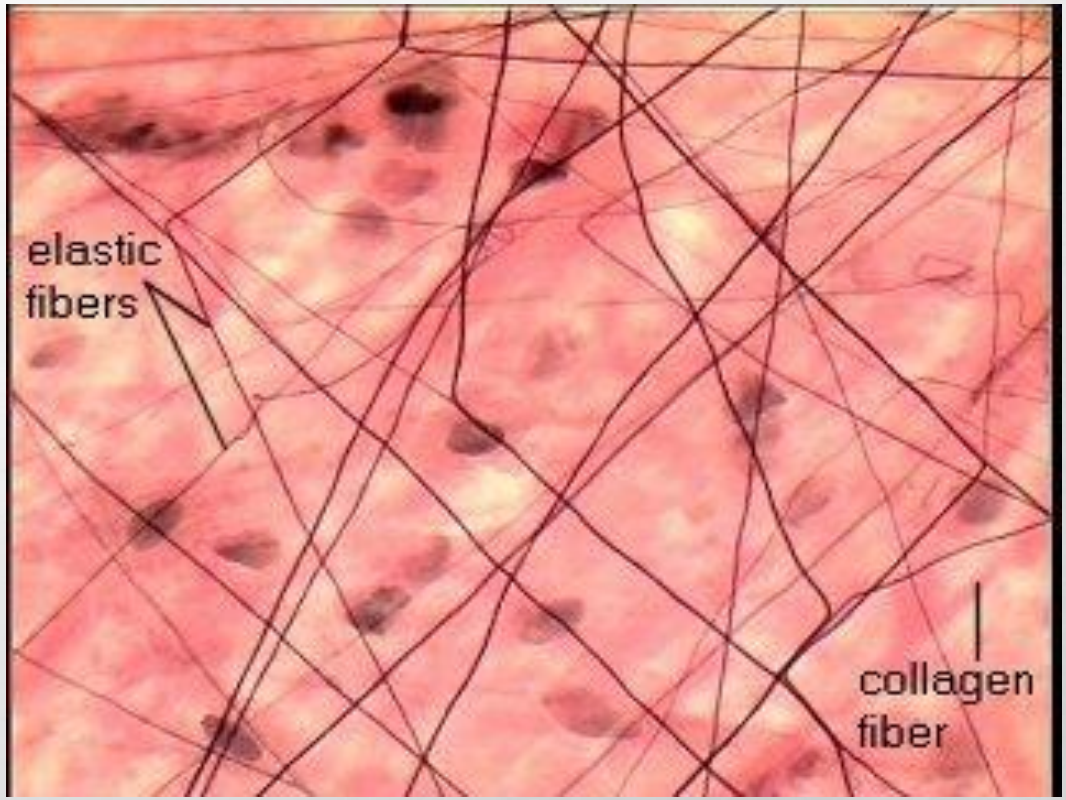
ELASTIC FORCES DUE THE FLUID LINING THE ALVEOLI

**TISSUE ELASTIC FORCES (A)** = REPRESENT 1/3 OF TOTAL LUNG ELASTICITY

**FLUID AIR SURFACE TENSION ELASTIC FORCES IN ALVEOLI (B)** = 2/3 OF TOTAL LUNG ELASTICITY.

# ELASTIC FORCES OF THE LUNG PARENCHYMA

Elastin and Collagen fibres are interwoven to form lung parenchyma



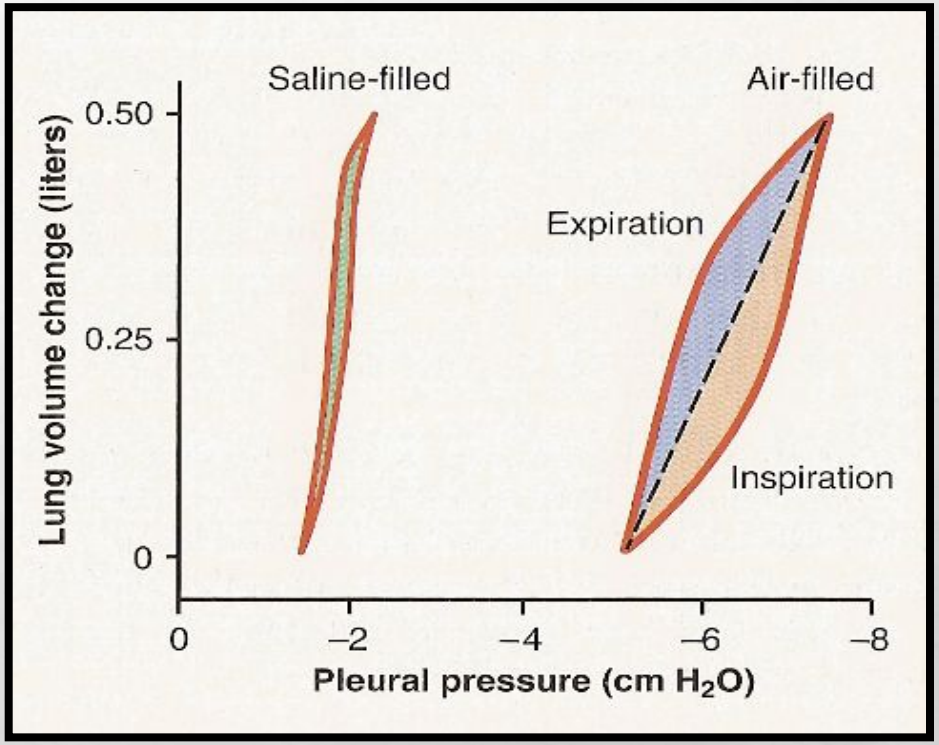
# ELASTICITY CAUSED BY SURFACE TENSION FORCES

- THE SURFACE OF THE ALVEOLI IS LINED BY A THIN LAYER OF FLUID
- THIS CONSTITUTES AN AIR-FLUID INTERFACE
- WHEN ALVEOLI ATTEMPT TO EXPAND, THESE SURFACE ACTIVE FORCES TRY TO OPPOSE AND COLLAPSE THE ALVEOLI



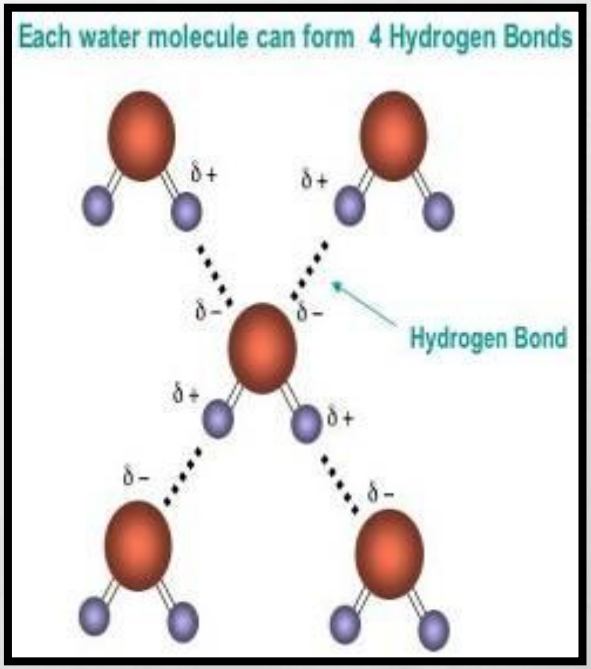
# COMPARISON OF LUNG COMPLIANCE IN AIR FILLED AND SALINE FILLED LUNG

- By adding saline solution there is no interface between air and alveolar fluid.
- Surface tension is not present, only elastic forces of tissue
- Transpleural pressures required to expand normal lung = 3x pressure to expand saline filled lung.



# ELASTIC FORCES OF LUNG PARENCHYMA

- **Water molecules are attracted to one another.**
- The force of surface tension acts in the plane of the air-liquid boundary to shrink or minimize the liquid-air interface
- In lungs = water tends to attract forcing air out of alveoli to bronchi = **alveoli tend to collapse (!!!)**




surface tension elastic force

⇩

Elastic contractile force of the entire lungs (forces B)

# PULMONARY SURFACTANT

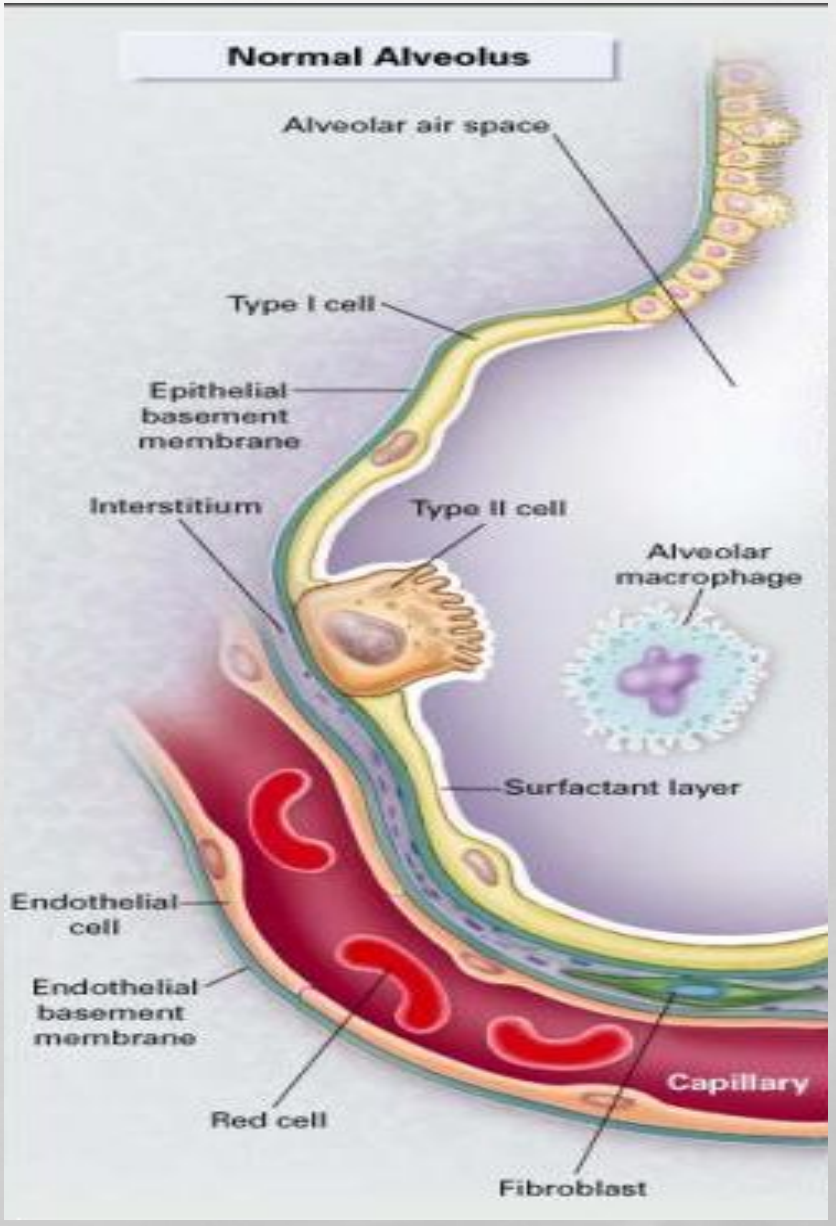
Surfactant is a surface acting material or agent that is responsible for lowering the surface tension of a fluid. Surfactant that lines the epithelium of the alveoli in lungs is known as pulmonary surfactant and it decreases the surface tension on the alveolar membrane.

Surfactant is a lipoprotein complex formed by lipids especially phospholipids, proteins and ions.

The most important components are the phospholipid dipalmitoyl phosphatidylcholine, surfactant apoproteins, and calcium ions.

# PULMONARY SURFACTANT

- SYNTHESIZED BY TYPE II ALVEOLAR CELLS
- **REDUCES SURFACE TENSION (PREVENTS ALVEOLAR COLLAPSE DURING EXPIRATION)**
- PREVENTS BACTERIAL INVASION
- CLEANS ALVEOLI SURFACE
- CONSISTS ON HYPOPHASE (PROTEIN) + PHOSPHOLIPID (DIPALMITOYLPHOSPHATIDYLCHOLINE) + CALCIUM IONS

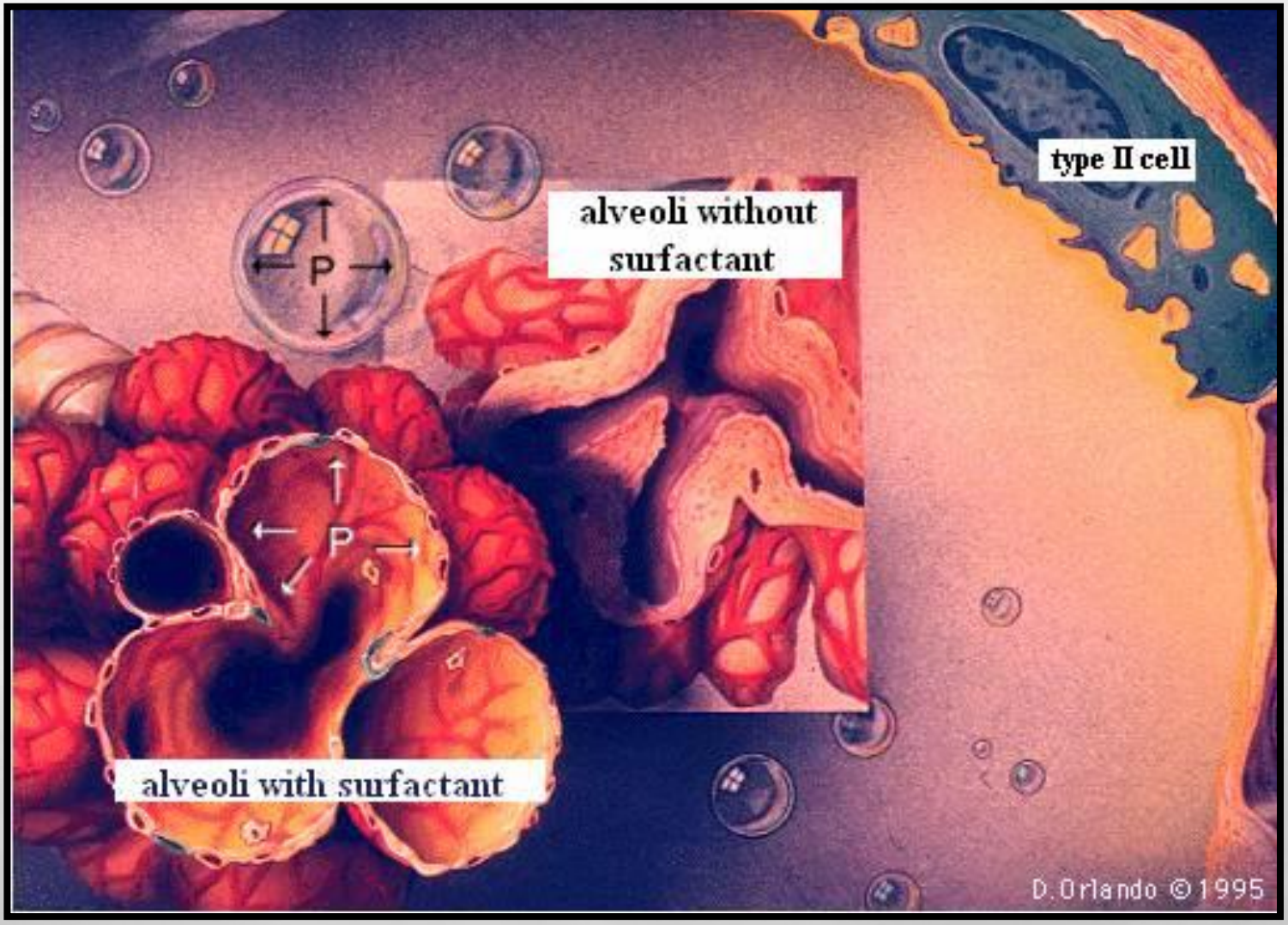


# ROLE OF SURFACTANT

Surface active agent in water = reduces surface tension of water on the alveolar walls

| | |
|---|---|
| Pure water (surface pressure) | 72 dynes/cm |
| Normal fluid lining alveoli **without** surfactant (surface pressure) | 50 dynes/cm |
| Normal fluid lining alveoli **with** surfactant | 5-30 dynes/cm |

# COLLAPSED ALVEOLI WITHOUT SURFACTANT



# LAPLACE LAW

$$\text{pressure} = \frac{(2 \times \text{surface tension})}{\text{radius}}$$

- "The pressure inside a balloon is calculated by twice the surface tension, divided by the radius."

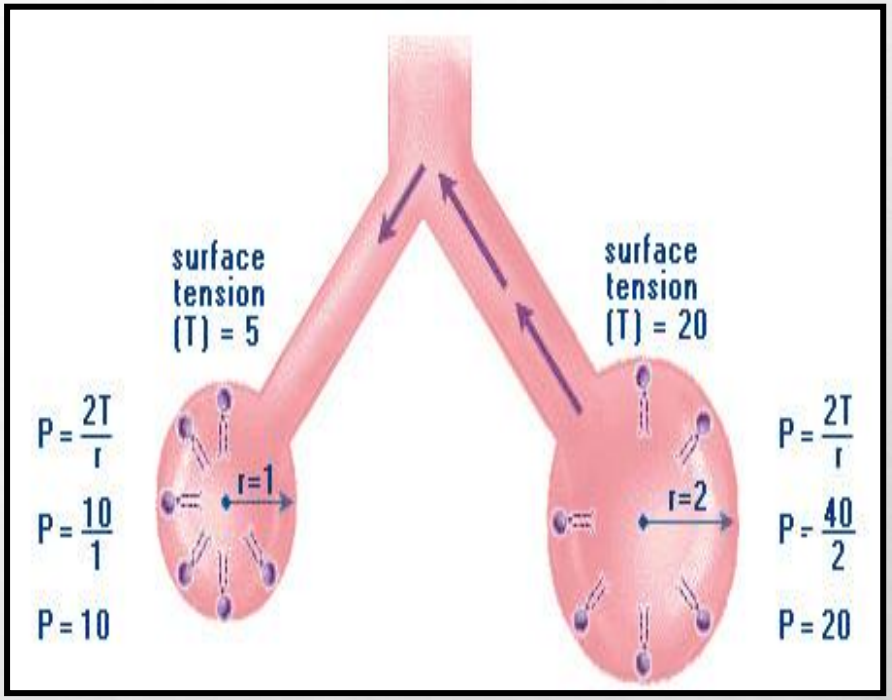
- Pressure to collapse generated by alveoli is inversely affected by radius of alveoli

↑ Radius = ↓ Pressure to collapse

- The smaller a bubble, the higher the pressure acting on the bubble
- **Smaller alveoli have greater tendency to collapse**

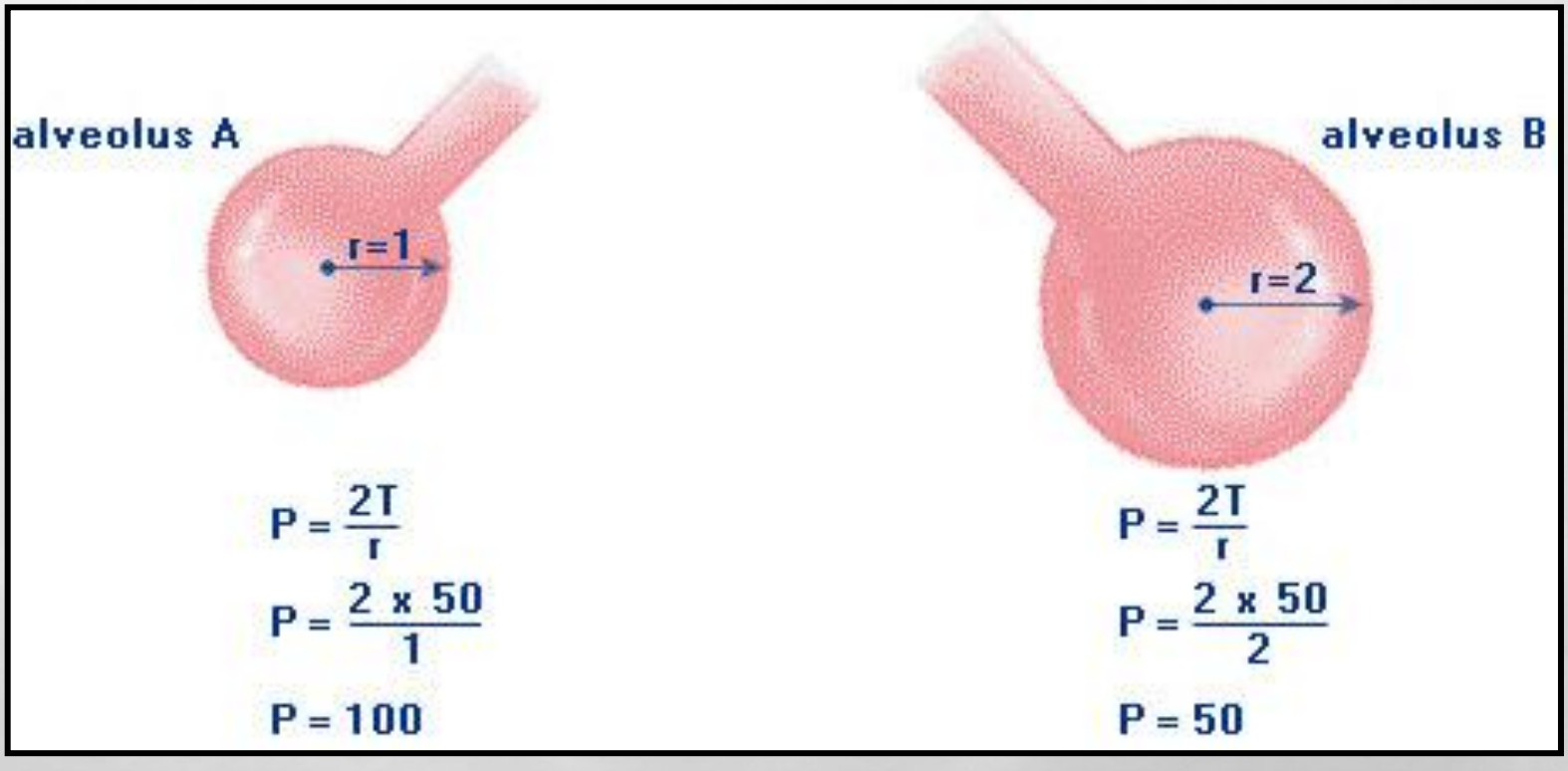
# SURFACE TENSION IN SMALL AND LARGE ALVEOLI

**If some alveoli were smaller and other large = smaller alveoli would tend to collapse and cause expansion of larger alveoli**



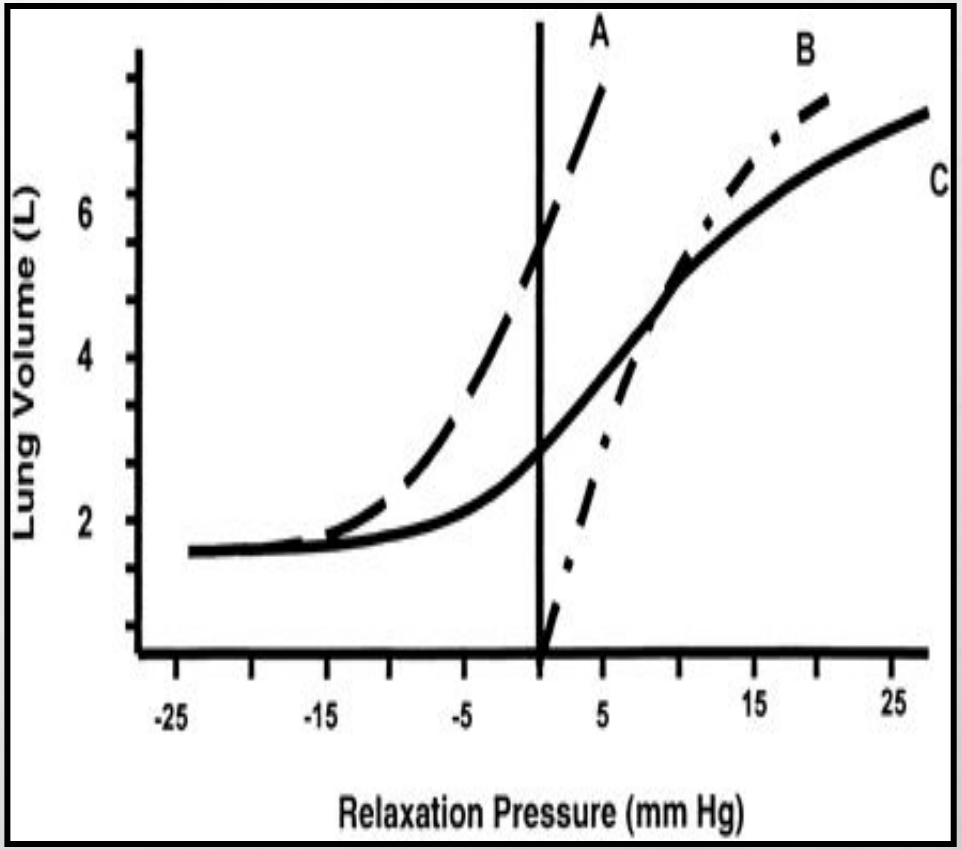

- This is prevented due to:
  - Normally larger alveoli do not exist adjacent to small alveoli = because they share the same septal walls.
  - All alveoli are surrounded by fibrous tissue septa that act as additional splints.
  - Surfactant reduces surface tension = as alveolus becomes smaller surfactant molecules are squeezed together increasing their concentration = reduces surface tension even more.

# EFFECT OF RADIUS ON SURFACE TENSION FORCES



# COMPLIANCE OF THORAX AND LUNG TOGETHER

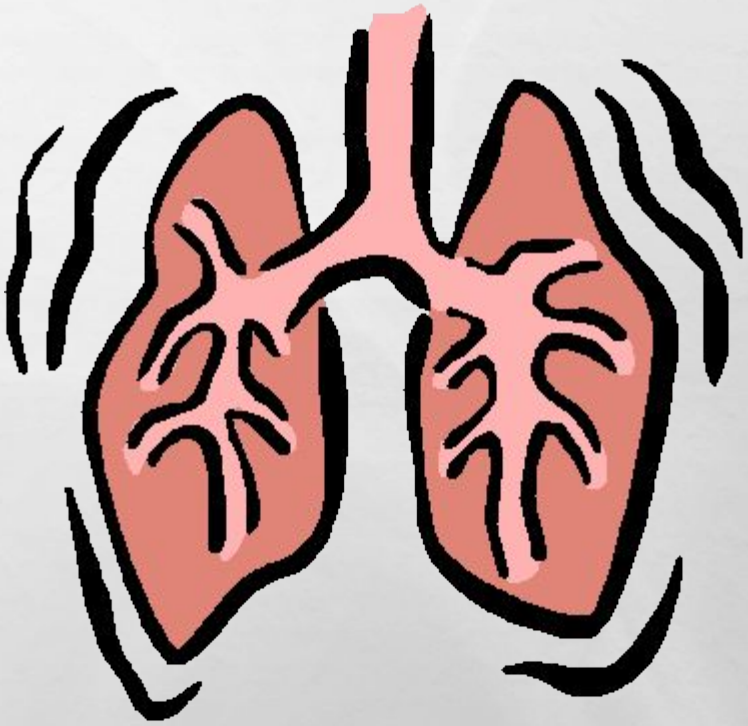
- Compliance of whole system is measured while expanding lungs of totally relaxed or paralysed person.
- Air is forced into the lungs a little at a time while recording lung pressures and volumes.
- The compliance of lungs and thorax = 1/2 of lungs alone.
- When lungs are expanded to high volumes or compressed to low volumes = limitations of chest wall increase = compliance of system is less than 1/5




chest cage (***A***), lung (***B***), combined chest lung cage (**C**)

# RESISTANCE IN AIR PASSAGES

- THE AIR PASSAGES ALSO PROVIDE RESISTANCE
- THERE MAY BE RESISTANCES IN THE PASSAGES FOR EXAMPLE MUCOUS PROVIDES A PHYSICAL BARRIER
- ALSO THE DIAMETER OF THE AIR PASSAGES PRODUCES RESISTANCE.
- FOR EXAMPLE A THINNER TUBE (AIR PASSAGE) PROVIDES MORE RESISTANCE THAN A TUBE WITH A LARGER DIAMETER
- THESE CAN PROVIDE RESISTANCE TO AIR WHEN YOU INSPIRE



# APPLIED PHYSIOLOGY

## Increase in Compliance

Compliance increases due to loss of elastic property of lung tissues, which occurs both in physiological and pathological conditions:

1. Physiological condition: Old age
2. Pathological condition: Emphysema

# Decrease in Compliance

Compliance decreases in several pathological conditions such as:

- Deformities of thorax like kyphosis and scoliosis
- Fibrotic pleurisy (inflammation of pleura resulting in fibrosis)
- Paralysis of respiratory muscles
- Pleural effusion
- Abnormal thorax such as pneumothorax, hydrothorax, hemothorax and pyothorax

# THANK YOU